



**LABMENTEX INTERNSHIP
PROJECT REPOT**

**Project title: DeepCSAT – Ecommerce
Customer Satisfaction Score Prediction**

Domain: AIML

Project type : Individual

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ABSTRACT

Customer satisfaction (CSAT) is a critical metric in e-commerce, directly influencing service quality, customer retention, and overall business growth. This project presents a predictive model for CSAT scores using a Deep Learning Artificial Neural Network (ANN). By leveraging customer interaction and support-related features, the model aims to generate actionable insights that help organizations identify satisfaction drivers and areas needing improvement. The ANN architecture is designed to capture complex, non-linear relationships within customer data, enabling more accurate predictions compared to traditional statistical methods. The outcomes of this project demonstrate the potential of deep learning in enhancing customer experience management, offering businesses a data-driven approach to improve service delivery and strengthen long-term customer relationships.

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INTRODUCTION

Customer satisfaction (CSAT) is one of the most vital indicators of business success in the digital era. In e-commerce, where customer interactions are largely virtual and competition is intense, CSAT serves as a direct reflection of how well a company meets customer expectations. High satisfaction levels not only foster loyalty and repeat purchases but also drive positive reputation, which is crucial in an environment where online reviews and ratings significantly influence consumer decisions. Conversely, poor satisfaction can quickly lead to customer attrition, negative publicity, and declining market share.

Understanding and predicting CSAT scores is therefore essential for organizations seeking to improve service quality, enhance customer retention, and sustain long-term growth. Traditional methods of measuring satisfaction often rely on surveys and feedback forms, which can be limited in scope and reactive in nature. With the rise of big data and advanced analytics, businesses now have the opportunity to proactively assess satisfaction by analyzing customer support interactions, behavioral patterns, and service features.

This project introduces a predictive model for CSAT scores using a Deep Learning Artificial Neural Network (ANN). The ANN architecture is particularly suited for capturing complex, non-linear relationships within customer data, enabling more accurate and nuanced predictions compared to conventional statistical approaches. By leveraging customer interaction and support-related features, the model aims to generate actionable insights that help organizations identify key drivers of satisfaction, anticipate dissatisfaction, and implement targeted improvements. Ultimately, this approach demonstrates how deep learning can transform customer experience management, offering businesses a powerful, data-driven tool to strengthen relationships and achieve sustainable growth.

DATASET

The data used in this project contains information about factors which influence customer satisfaction. These attributes provide a holistic view of how customers perceive their interactions with the platform. By analyzing these features, the model can identify patterns that distinguish satisfied customers from dissatisfied ones. This comprehensive dataset enables the Artificial Neural Network to learn complex relationships and generate accurate predictions of CSAT scores.

Attributes in the dataset

Unique id	Unique identifier for each record
Channel name	Name of the customer service channel
Category	Category of the interaction
Sub-category	<u>Sub-category</u> of the interaction
Customer Remarks	Feedback provided by the customer
Order id	Identifier for the order associated with the interaction
Order date time	Date and time of the order
Issue reported at	Timestamp when the issue was reported
Issue responded	Timestamp when the issue was responded to
Survey response date	Date of the customer survey response
Customer city	City of the customer
Product category	Category of the product
Item price	Price of the item

Connected handling time	Time taken to handle the interaction
Agent name	Name of the customer service agent
Supervisor	Name of the supervisor
Manager	Name of the manager
Tenure Bucket	<u>Bucket categorizing agent tenure</u>
Agent Shift	Shift timing of the agent
CSAT Score	Customer Satisfaction (CSAT) score

Graphical Representation of Feature Interactions

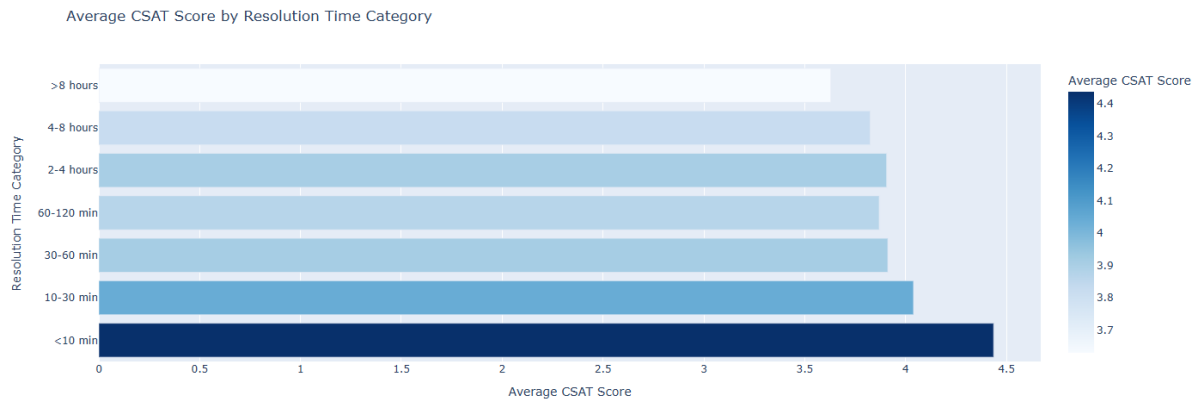


Fig 1 : Chart showing how CSAT scores vary with the time taken to respond to customer issues.

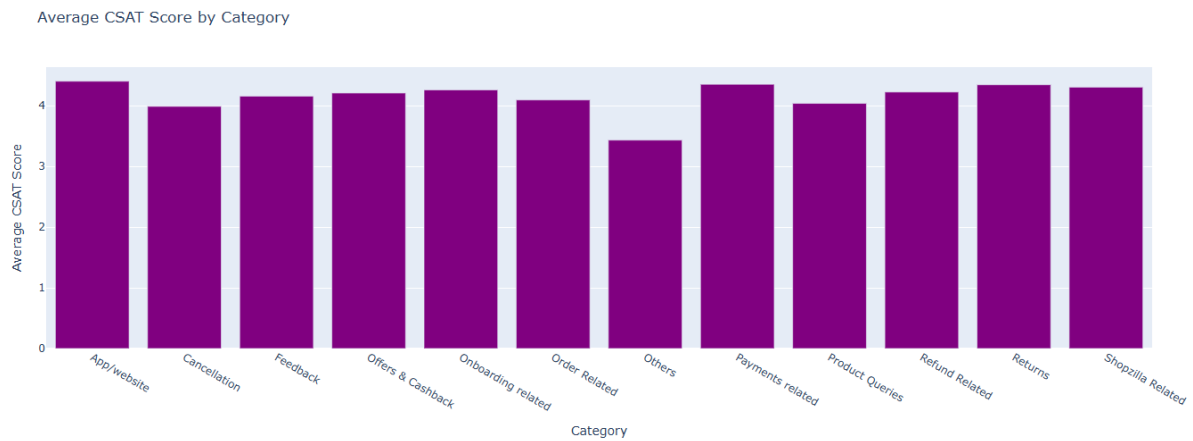


Fig 2 : Chart showing the average CSAT scores for each category of the issue resolved

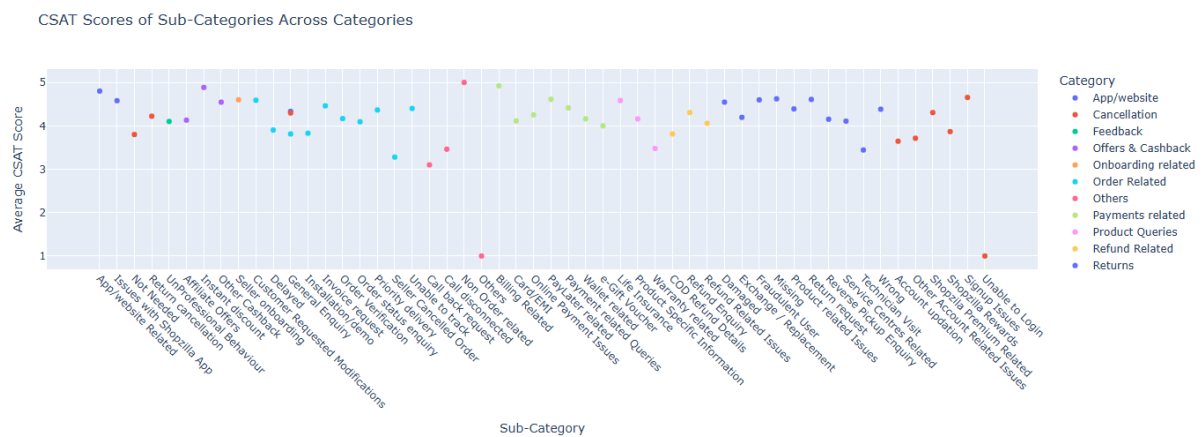


Fig 3 : A scatter plot representing CSAT scores of each sub-category within a category



Fig 4 : Chart helps representing the average CSAT score for each type of channel used as a medium to resolve the issue.



Fig 5 : The above chart represents CSAT scores distribution across different agent shift.

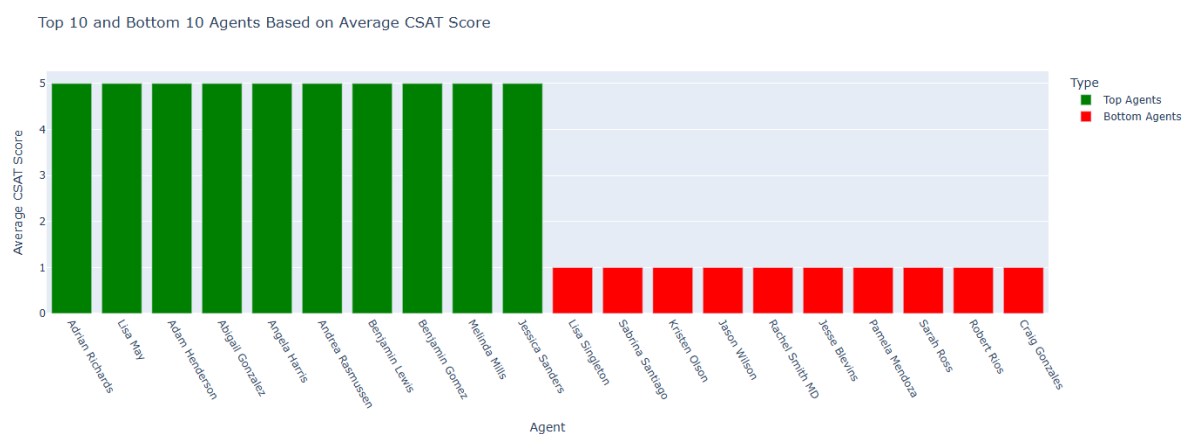


Fig 6 : Analysing CSAT Scores based on agent performance.

METHODOLOGY

1.Data preprocessing

1.1 Handling missing values

The dataset contains few columns with large number of missing values, few columns with moderate missing values and few columns with very few missing values. Columns containing more than 70% missing values are removed from the dataset. For columns containing around lesser number of missing data, the missing values are replaced with standard values such as "not provided" for text data and statistical values such as mean, median or mode of the values in that column.

1.2 Handling outliers

Outliers are values in the dataset which differ largely from the mean or average values of the data. This may introduce noise in the data which affects the training of the model. Therefore handling outliers is an important step in data preprocessing.

IQR technique - This method finds values that are far away from the middle 50% of the data. Any data less than 25th percentile or greater than 75th percentile of the data is considered as an outlier. *Winsorization* - These outliers are capped to the boundary values(either 25% or 75%) for data analization.

1.3 Categorical encoding

Categorical encoding converts categorical data into numerical values such that it can be easily interpreted by the machine learning model.

One-hot encoding - Encoding categorical values into binary values for each category. This technique best suited for smaller number of categories. Therefore, categories such as channel name, agent shift, category and sub-category are encoded using one-hot encoding.

Label encoding - This technique converts categorical data into numerical labels so that the model can process them. It is best suited for categories with many categorical values like variables such as agent, supervisor, managers which are encoded using this technique.

Ordinal encoding - This method is useful for ordered data such as tenure buckets. It groups these data into numerical values. Hence categorical column such as tenure buckets is encoded into numerical values using this technique.

1.4 Feature engineering

This step in preprocessing involves removal of highly correlated features as it might lead to data redundancy which affects the model training. A correlation heatmap is used to determine the correlation co-efficient between the variables. The variables having correlation greater than 0.9 are removed from the dataset. Also, new features which are useful in determining factors that influence the customer satisfaction are added to the dataset such as response time which is calculated by subtracting issue response time and issue reported time and agent frequency which represents number of tasks handled by each agent.

1.5 Feature selection

SelectKbest - It is a statistical feature selection method which is applied for selecting top k features which mostly influence the customer satisfaction scores. This technique tests each feature against the target variable(CSAT score). After evaluating each feature, statistical test is performed for every feature to calculate F-score. F-score measures how each feature is linearly related to target variable. These scores are then sorted to find the top k features with highest F-scores.

1.6 Data splitting

The data is split into training and testing data. Training data is used to train the machine learning model where it learns patterns, trends and relationship from the dataset. Testing data is used to evaluate the trained model. The model performs predictions on the test data using the insights gained from training data. The data is split into 80% training data and 20% as test data. This ratio provided enough data for the model to learn while maintaining sufficient data for testing.

1.7 Data balancing

From the dataset, it can be inferred that the CSAT score distribution is imbalanced, with the value 5 occurring most frequently as the majority class, while scores such as 2.5, 3, and 4 appear less often as minority classes. This imbalance may introduce bias in the model during training, as the model may tend to predict the majority class more frequently. To address this issue, the dataset is balanced using the SMOTE technique, which generates synthetic samples for the minority classes. This helps create a more balanced distribution, enabling the model to learn patterns from all classes more effectively and improve prediction performance.

1.8 Data scaling

StandardScaler - It is a data scaling technique used to standardize numerical features so that they have a mean of 0 and a standard deviation of 1. This transformation ensures that all features are placed on a similar scale, which is particularly important for machine learning algorithms such as Artificial Neural Networks that rely on gradient-based optimization. Without scaling, features with larger numerical values may dominate the learning process and influence the model more than features with smaller values. By standardizing the data, StandardScaler helps the model treat all input variables more equally. This improves the stability of the training process, accelerates convergence during optimization, and leads to more efficient and reliable model performance.

2. Building and training the model

The model is built using an Artificial Neural Network (ANN) with the help of TensorFlow and Keras. The input layer receives the preprocessed and scaled features such as response time, category, agent details, and channel information. The network contains several dense (fully connected) layers with ReLU activation, which helps the model learn complex patterns between these features and the CSAT score. Batch Normalization layers are added to stabilize and speed up training, while Dropout layers reduce overfitting by randomly disabling some neurons during training. The final layer uses a Softmax activation function with 5 neurons to predict the probability of each CSAT class (scores 1–5).

The model is trained using the Adam optimizer, which automatically adjusts learning rates to improve convergence. The Sparse Categorical Crossentropy loss function is used because the target variable is a multi-class classification problem with integer labels. During training, class weights are applied to handle any class imbalance so that the model learns all CSAT classes fairly. Early Stopping monitors validation loss and stops training when the model stops improving, preventing overfitting and reducing unnecessary training epochs. The model is trained on the balanced and scaled dataset, and its performance is evaluated on the test set using accuracy and other classification metrics.

3 Hyperparameter tuning

The hyperparameter optimization technique used in this implementation is Random Search, performed using Keras Tuner (kt.RandomSearch). Random Search works by randomly sampling combinations of hyperparameters—such as the number of neurons in hidden layers (units1, units2), dropout rates (dropout1, dropout2), and the learning rate from predefined ranges and evaluating each configuration based on validation accuracy. In this model, max_trials=5 determines the number of different hyperparameter combinations tested during tuning. Random Search was chosen because it is computationally more efficient than exhaustive methods like Grid Search, especially when the search space is large. Instead of evaluating every possible combination, it explores diverse regions of the hyperparameter space

and can often identify near-optimal configurations faster, making it suitable for tuning deep learning models such as the ANN used for CSAT score prediction.

4 Evaluation

Model Accuracy: 0.6759981375858457

Classification Report:

	precision	recall	f1-score	support
2	0.14	1.00	0.25	2468
3	0.00	0.00	0.00	503
4	0.00	0.00	0.00	2277
5	0.00	0.00	0.00	11934
accuracy			0.14	17182
macro avg	0.04	0.25	0.06	17182
weighted avg	0.02	0.14	0.04	17182

Confusion Matrix:

```
[[ 275   52   0 2141]
 [   38    9   0  456]
 [   89   48   0 2140]
 [  364  239   0 11331]]
```

5 Deployment

The model is deployed using streamlit platform to create a frontend which takes response time, category, sub-category, agent name, channel name, agent shift, manager and agent tenure bucket as input and predicting the CSAT score as output. Streamlit is an open-source Python framework used to build interactive web applications for data science and machine learning models. The app can be executed or deployed locally using the system terminal.

Commands to run the app

```
>> pip install -r requirements.txt
```

```
>> pip install streamlit
```

```
>> streamlit run app.py
```

Github link - <https://github.com/AkhilaRao0002/DeepCSAT-Ecommerce-Customer-Satisfaction-Score-Prediction>

Customer Satisfaction (CSAT) Prediction

Enter the support interaction details to predict CSAT score.

Response Time

30.00

- +

Category

Product Queries

Sub Category

Life Insurance

Agent Name

Richard Buchanan

Channel Name

Outcall

Agent Shift

Morning

Manager

Jennifer Nguyen

Tenure Bucket

On Job Training

Predict CSAT Score

Predicted CSAT Score: 5

Fig 7 : Screenshot of frontend for model deployment

CONCLUSION

This project focused on predicting Customer Satisfaction (CSAT) scores using machine learning techniques. After performing data preprocessing, feature selection, balancing, and scaling, an Artificial Neural Network (ANN) model was developed to learn patterns between customer support features and satisfaction scores. Hyperparameter tuning and regularization techniques such as dropout and batch normalization were applied to improve model performance and generalization.

The final model was able to predict CSAT scores effectively, helping identify important factors that influence customer satisfaction, such as response time and issue category. Additionally, the model was deployed using Streamlit to create an interactive interface where users can input data and obtain CSAT predictions.

Overall, this system can help organizations monitor customer experience, identify service issues, and make data-driven decisions to improve customer support quality and satisfaction.